

A Hybrid Deep Learning Approach With Explainable AI for Diabetic Retinopathy Classification

Paper Critique & Group Technical Evaluation

Research Review Group

Academic Review

September 24, 2026

Executive Summary & Clinical Context

- **Global Burden:** Diabetic Retinopathy (DR) affects > 150 million diabetes patients worldwide.
- **Clinical Challenge:** Leading cause of blindness among working-age adults.
- **Screening Bottleneck:** Manual ophthalmic fundus inspection is time-consuming and expensive.

Identified Research Gaps

- **Severe Class Imbalance:** APTOS 2019 dataset heavily skewed toward No DR (49%).
- **Single-Backbone Limits:** Standard CNNs fail to capture both macro patterns and fine lesions.
- **Black-Box AI:** Lack of quantitative explainability hinders clinical adoption.

Core Research Objectives

- **5-Stage Grading:** Classify fundus images across No DR, Mild, Moderate, Severe, Proliferative DR.
- **Feature Fusion:** Combine EfficientNetB0 (multi-scale) and DenseNet121 (dense connections).
- **Quantitative XAI:** Introduce Grad-CAM concentration score to measure diagnostic focus.

- **Dataset:** APTOS 2019 Blindness Detection (3,662 fundus images).
- **Split Ratio:** 70% Training (2,562), 20% Validation (733), 10% Testing (367).
- **Normalization:** Resized to 224×224 pixels with ImageNet mean/std scaling.

SMOTE Class Balancing Strategy

- **Resampling Target:** Balanced training classes to 1,263 samples each (6,315 total).
- **Pixel-Level Oversampling:** Applied SMOTE on normalized pixel space representations.
- **Evaluation Guardrail:** Test set (367 images) kept 100% untouched to avoid data leakage.

Evaluated Baseline CNN Backbones

- **ResNet-18:** Residual shortcuts (11.18M parameters).
- **EfficientNet-B0:** Compound scaling with MBConv blocks (4.01M parameters).
- **DenseNet-121:** Dense feature reuse for micro-lesions (6.96M parameters).
- **MobileNet-V2:** Lightweight inverted bottlenecks (2.23M parameters).

Proposed Hybrid Fusion Architecture

- **Feature Extractor:** Concatenated DenseNet-121 (1024-D) + EfficientNet-B0 (1280-D).
- **Unified Vector:** 2304-dimensional fused representation.
- **Classifier Head:** Linear (2304 $\rightarrow$ 512) $\rightarrow$ ReLU $\rightarrow$ Dropout (0.4) $\rightarrow$ Linear (512 $\rightarrow$ 5).

Ensemble Learning Strategies

- **Hard Voting:** Majority decision over discrete backbone class predictions.
- **Soft Voting:** Aggregated average of softmax class probability distributions.
- **Advantage:** Reduces model-specific variance across ambiguous severity boundaries.

Experimental Performance Comparison

Table: Model Performance Summary

Model	Accuracy	Precision	Recall	F1-Score
ResNet-18	85.01%	84.90%	85.01%	84.39%
EfficientNet-B0	83.65%	83.22%	83.65%	82.81%
DenseNet-121	82.83%	82.27%	82.83%	82.12%
MobileNet-V2	83.38%	83.04%	83.38%	82.95%
Ensemble (Soft)	85.29%	84.49%	85.29%	84.58%
Hybrid Model	84.02%	83.12%	84.02%	82.91%

Explainable AI & Grad-CAM Findings

- **Visual Heatmaps:** Confirmed model attention on microaneurysms and exudates.
- **Concentration Score:** Hybrid EfficientNet branch achieved 0.6586 vs DenseNet 0.4244.
- **Localization:** Hybrid features showed tighter spatial focus on retinal lesions.

Table: Hardware Complexity Metrics

Model	Params (M)	GFLOPs	Inference (ms)
MobileNet-V2	2.23	0.33	8.49
EfficientNet-B0	4.01	0.41	12.63
DenseNet-121	6.96	2.90	24.85
ResNet-18	11.18	1.82	3.60
Hybrid Model	14.45	19.86	High (Batch)

Critical Paper Strengths

- **Quantitative XAI:** Moved beyond qualitative heatmaps using concentration scores.
- **Systematic Comparison:** Evaluated individual, ensemble, and hybrid methods under identical splits.
- **Per-Class AUC:** High discriminative power across stages (> 0.90 ROC-AUC).

Methodological Flaws & Weaknesses

- **Single Dataset:** Lacks cross-dataset validation on Messidor-2 or EyePACS.
- **Pixel SMOTE:** Applying SMOTE on flattened images creates artificial spatial interpolation.
- **Efficiency Paradox:** Hybrid model FLOPs expanded (19.86 GFLOPs) without exceeding ResNet-18 accuracy (85.01%).

- **Latent Feature Oversampling:** Apply SMOTE on feature embeddings rather than raw pixel vectors.
- **External Cohort Validation:** Evaluate generalization on external clinical fundus datasets.
- **Model Compression:** Apply knowledge distillation to compress the 2304-D classifier head.

Concluding Verdict

- Solid contribution to quantitative XAI and feature fusion benchmarking.
- Methodologically limited by single-dataset scope and raw pixel oversampling.
- **Final Rating:** Acceptable baseline study with clear avenues for future extension.